

MindfulReframer: A Virtual Reality System for Managing Exam Anxiety Through Cognitive Reappraisal

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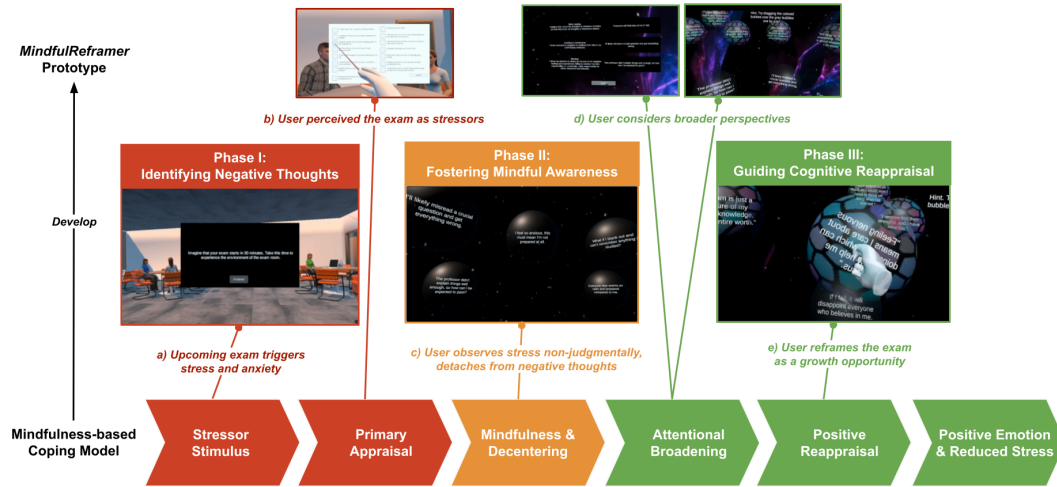


Figure 1: A summary of how the MindfulReframer system integrates Mindfulness-Based Cognitive Therapy (MBCT) principles – see the Mindfulness-based Coping Model for detail (adapted from Garland et al. [12]) – to facilitate Cognitive Reappraisal (CR) through VR, helping the user manage negative thoughts and emotions triggered by an exam scenario. The MindfulReframer system incorporates three key phases: (i) *Identifying Negative Thoughts*, where users reflect on personal stressors in a virtual exam environment and identify three specific negative thoughts; (ii) *Fostering Mindful Awareness*, which immerses users in an abstract “thinking space” to engage non-judgmentally with their thoughts and emotions through interactive visualizations and guided mindfulness exercises; and (iii) *Guiding Cognitive Reappraisal*, where users are guided in reframing negative thoughts through embodied interaction with their negative thoughts.

Abstract

Enhancing emotion regulation (ER) among university students is essential for improving their academic performance and psychological well-being. MindfulReframer, a novel Virtual Reality (VR) system, represents the first interactive tool to help students manage exam-related anxiety by teaching Cognitive Reappraisal (CR) strategies through Mindfulness-Based Cognitive Therapy (MBCT). Immersing participants in realistic exam scenarios guides them to identify negative thoughts, foster mindful awareness, and practice CR in an interactive thought space. A mixed-methods evaluation showed significant improvements in reappraising stressful thoughts and regulating emotions ($p = .003$). The metaphor-driven, immersive

design externalizes emotions, promotes non-judgmental awareness, and enhances cognitive flexibility, supporting better control over negative emotions and effective CR. MindfulReframer addresses a critical gap in mental health interventions for academic stress management and everyday emotion regulation.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Empirical studies in HCI**.

Keywords

Virtual Reality; Emotion Regulation; Cognitive Reappraisal; Exam Anxiety; Mindfulness-Based Cognitive Therapy

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1 Introduction

Emotions are central to human behaviour, stemming from complex behavioural and physiological responses to environmental stimuli and shaping social interactions [17]. Anxiety is notably prevalent among university students, with up to 74% experiencing clinically significant levels [26]. Characterized by heightened arousal, negative valence, and reduced control, anxiety impairs cognitive functions like memory, attention, and decision-making [6]. In this context, emotion regulation (ER) – the ability to manage and respond to emotional experiences in adaptive ways – becomes essential in everyday life [33].

In universities, academic stressors such as exams, presentations, and social interactions often intensify anxiety, affecting academic performance and psychological well-being [6, 27]. Poor ER leads to maladaptive behaviours that provide temporary relief but long-term distress [27]. Thus, equipping students with adaptive ER skills like Cognitive Reappraisal (CR) – reframing stress-inducing events to reduce emotional impact – is crucial [9]. CR has been widely recognised for its effectiveness in reducing negative emotions, increasing positive emotions, and improving overall performance across various contexts [9, 11, 19]. While effective, CR can be challenging to learn and apply without appropriate guidance [9, 10, 34].

Virtual Reality (VR) has emerged as a promising tool for supporting ER due to its unique capacity to create immersive environments that foster a deep sense of presence and engagement [8]. Studies have demonstrated VR's potential to elicit real-time, authentic emotional responses by creating realistic and personally relevant environments [20], making it particularly suited for ER interventions [2, 7]. While VR has been successfully applied in clinical settings, such as exposure therapy for anxiety disorders [7], its application in supporting the emotional well-being of healthy individuals remains underexplored. Recent studies have begun to explore VR's potential to help healthy individuals manage negative emotions through visual representations of thoughts and emotions in immersive environments [30, 38]. However, these studies mainly focus on general negative emotions in everyday context. Significant gaps remain in how VR can support specific adaptive ER strategies, such as CR, particularly in the context of academic stress experienced by university students.

Building on these insights, we introduce MindfulReframer, a VR system designed to help university students develop CR strategies to manage exam-related anxiety. Our VR system is grounded in Mindfulness-Based Cognitive Therapy (MBCT) [21], encouraging present-moment awareness and the non-judgmental observation of thoughts and feelings, which is essential in facilitating CR [12, 37]. By integrating MBCT principles with VR technology, MindfulReframer immerses users in a virtual environment where they can practice isolating themselves from negative thoughts and engage in CR guidance. This VR system provides university students with a practical, interactive tool for managing the stress and anxiety that typically accompany high-stakes academic evaluations, such as exams.

The present study aims to design and implement a novel VR system, MindfulReframer, to facilitate CR strategies for managing exam-related anxiety. We evaluated the efficacy of MindfulReframer

using qualitative and quantitative methods with 22 university students. Our findings indicate that MindfulReframer significantly improved participants' CR skills ($p = 0.003$) while fostering positive emotional responses and enhanced emotional control. These results suggest that MindfulReframer is an effective tool for helping students manage the negative emotions and thoughts associated with exam anxiety through mindfulness and CR techniques.

This research makes two main contributions: (1) the design and implementation of MindfulReframer – a VR system developed based on MBCT to facilitate CR and enhance ER in response to exam-related anxiety; (2) a mixed-method study that evaluates the effectiveness of MindfulReframer in supporting ER through mindfulness and interactive metaphors.

2 Related Works

In this section, we briefly review the role of VR in supporting ER and introduce MBCT as a foundation for enhancing CR through VR.

2.1 The Potential of Virtual Reality for Facilitating Emotion Regulation

VR technology can create immersive, interactive environments that evoke genuine emotional responses [20], prompting research into its potential to support Emotion Regulation (ER) [15]. For instance, one study found that a peaceful VR forest with a mindfulness voiceover significantly increased state mindfulness and positive affect in the general population [31]. Similarly, *Moodshaper*, a VR tool for managing negative emotions, improved positive affect and reduced ER difficulties and negative affect [38]. *Mind Mansion*, another VR application, immerses users in a cluttered virtual apartment to promote thought awareness and mental self-care, reducing negative affect and enhancing emotional awareness [30]. While these studies highlight VR's potential in ER, they mainly focus on general negative emotions, leaving a gap in understanding how VR can support targeted strategies like CR in addressing specific emotions, such as academic anxiety.

2.2 Cognitive Reappraisal as an Adaptive Emotion Regulation Strategy

According to Gross's process model of ER, strategies that work in the early stages of emotional processing – before the emotional response is fully activated – are known as *antecedent strategies* [15]. One of the most widely researched antecedent-focused strategies is CR [11, 24, 40], which involves reinterpreting negative emotional situations as positive ones and changing the meaning assigned to the stimuli [14]. Compared to response-focused strategies, such as expressive suppression (i.e., inhibiting behaviours associated with emotional responses), CR has been found to be more effective in reducing negative emotions and enhancing positive emotion experience [15, 36]. However, despite its documented benefits, CR remains underutilised, particularly among university students [4, 5]. A potential reason for this underutilization is that CR requires a high level of cognitive resources and effort [28], making it challenging to implement effectively without additional support.

2.3 Integrating Mindfulness-Based Cognitive Therapy to Enhance Cognitive Reappraisal

Many theories now support VR-based ER, including artistic emotional expression and metaphors [30, 38]. MBCT combines cognitive-behavioural strategies with mindfulness practices to foster adaptive ER strategies, particularly CR [12, 37]. Mindfulness involves focusing attention on the present moment through meditation and other training [22]. It is hypothesized that mindful thinking helps shift attention away from evaluative language, promoting non-judgmental metacognition, or “decentering,” where one observes thoughts without attachment [21].

Two core MBCT principles support CR. First, non-judgmental acceptance fosters a decentered state, enhancing cognitive flexibility and helping disengage from initial emotional reactions, thus enabling cognitive reconstruction – replacing negative interpretations with positive ones [37]. Second, present-centred awareness helps individuals quickly recognize negative thoughts and emotions, providing a foundation for effective reappraisal [16]. Improved awareness also enhances the ability to reappraise challenging situations by being more conscious of one’s current state [37].

As shown in Figure 1, by grounding the VR experience in MBCT, we can design the process to support university students in regulating negative emotions or thoughts related to exams through two key steps:

- (1) By **adopting meditation**, students can detach themselves from immediate emotional reactions and broaden their range of attention, allowing them to generate a metacognitive perspective to detach themselves from stressful appraisals.
- (2) By **reappraising** stressful exams from a more constructive and growth-oriented perspective, such as interpreting exams as opportunities for learning and development rather than as threats, thereby reducing anxiety and fostering positive emotions.

Through this framework, we aim to equip students with adaptive ER skills, enabling them to manage academic stress and contribute to their long-term emotional well-being.

3 System Design

Building upon MBCT, MindfulReframer is developed to reduce users’ anxiety through three key phases: (i) *Identifying Negative Thoughts*, (ii) *Fostering Mindful Awareness*, and (iii) *Guiding Cognitive Reappraisal* (see Figure 1). The following section outlines the design principles for each stage, illustrating the scenario design. The system was developed using Unity version 2022.3.38 and operates on the Oculus Meta 3 device.

3.1 Phase I: Identifying Negative Thoughts

Phase I of MindfulReframer is designed to trigger negative emotions and identify negative thoughts related to exams. As shown in Figure 1, users are immersed in a virtual exam environment and prompted to imagine being 10 minutes from the exam’s start, which helps to induce realistic exam-related anxiety responses. Then, users are prompted to select three out of the fifteen negative thoughts that are most relevant to their personal experience (i.e. see Figure 1 b)(see Appendix A for the negative thoughts). These three

thoughts will serve as the primary material for the subsequent CR in phase III.

3.2 Phase II: Foster Mindful Awareness

In Phase II, users explore their relationship with negative exam-related thoughts through positive guidance. They enter an abstract “thinking space” with 15 negative thoughts as floating bubbles (i.e. see Figure 1 c). Users can grasp, throw, or release these bubbles, metaphorically managing thoughts and emotions. This design aligns with MBCT, viewing thoughts as transient mental events rather than objective realities [21]. Phase II includes a 4-minute guided mindfulness meditation from *The Mindful Way Workbook* [35], delivered via a gentle AI voice¹. The script encourages awareness of thoughts, emotions, and bodily sensations, enhancing the user’s ability to decenter, a precursor for effective CR [12].

3.3 Phase III: Guiding Cognitive Reappraisal Strategy

Phase III of MindfulReframer focuses on psychoeducation and guided CR strategies (i.e. see Figure 1 d). Users are guided to learn about the cognitive distortions underlying the three negative thoughts identified in Phase I (see Appendix A for the cognitive distortions). This process encourages users to broaden understanding of their general cognitive patterns, providing a foundation for the reappraisal process. Subsequently, they will be guided to explore alternative perspectives on their negative thoughts. In the VR environment, grey bubbles represent negative thoughts, and coloured bubbles represent healthier thoughts. Users select and place coloured bubbles onto the grey ones, promoting a shift in perspective (i.e. see Figure 1 e). This phase aligns with MBCT principles, encouraging users to change their relationship with thoughts and supporting cognitive restructuring for more constructive thinking [12].

4 Evaluation

The evaluation of MindfulReframer aimed to assess its effectiveness in upregulating emotion and enhancing CR by integrating quantitative data from questionnaires with qualitative insights from semi-structured interviews. The study received ethics approval from the Monash University Human Research Ethics Committee (MUHREC) (Approval No. 38321), and all data were collected and stored in accordance with institutional guidelines and ethical standards.

4.1 Participants

We recruited 22 undergraduate and graduate students (13 females, 9 males; $M = 23.65$ years, $SD = 1.23$) from diverse academic backgrounds via university mailing lists and campus flyers. Our inclusion criteria were as follows: 1) enrollment in a university program; 2) no history of diagnosed mental health disorders or photosensitivity. Among the participants, 6 had no prior experience with VR, 12 were sporadic users, and 4 were experienced VR users. After completing the experiment, participants were given a gift card worth US\$15 in local currency as compensation.

¹<https://artlist.io,byavoiceovercalledSerenity>

4.2 Measures

In the following, we report the details of the quantitative and qualitative measures collected in our experiment.

4.2.1 Emotion Regulation Questionnaire (ERQ). The ERQ is a widely used tool for assessing CR with six items reflecting the individuals' use of reappraisal in general (e.g., I control my emotions by changing the way I think about the situation I'm in). Responses are measured on a Likert scale from 1 (strongly disagree) to 7 (strongly agree) [15].

4.2.2 Self Assessment Manikin (SAM). The SAM is a well-established tool for assessing subjective emotional experiences based on a three-dimensional model of emotion: 1) *valence* (negative to positive), 2) *arousal* (calm to excited), and 3) *dominance* (lack of control to full control). The SAM uses graphic figures on a 9-point scale to allow participants to identify their emotional state in response to specific stimuli [3].

4.2.3 Interview. We conducted a semi-structured interview after the VR experience, with an average duration of eight minutes. During the interview, participants were asked about their overall impressions of the experiment, preferred and least preferred interactions, the reasons behind their preferences, possible mood shifts, and takeaways from the experience.

4.3 Procedure

Participants first submitted a signed informed consent form, followed by a briefing on the experimental procedure, as outlined in Figure 2. To begin, participants were asked to complete the ERQ to establish a baseline measure of their CR strategy propensities, followed by the SAM to assess their current emotional states before engaging with the VR environment. Participants then engaged in the three main phases of the MindfulReframer VR experience: (i) *Identifying Negative Thoughts*, (ii) *Fostering Mindful Awareness*, and (iii) *Guiding CR*. At the end of Phase I, participants completed the SAM again to assess their emotional state within the VR exam scenario. After the VR session, participants completed the SAM and ERQ again to measure any immediate changes in emotional states and ER strategies. The procedure concluded with a semi-structured interview.

4.4 Data Analysis

The quantitative data were analysed using R. Initially, the normality of the data was assessed using the Shapiro–Wilk test. Given that the data did not follow a normal distribution, non-parametric tests were employed. We used Wilcoxon signed-rank tests to examine whether there were significant differences in CR scores between the *pre* and *post* measurement. For the SAM data, Friedman tests were employed to examine changes across the three-time points (*pre*, *during* and *post* measurement) for each of the three dimensions: *valence*, *arousal*, and *dominance*. For qualitative analysis, all audio recordings were transcribed verbatim for thematic analysis.

5 Findings

This section presents quantitative results from the questionnaires and qualitative insights generated from the interviews.

5.1 Quantitative Findings

Wilcoxon Signed-Rank test was conducted to compare *pre* and *post* measurements for reappraisal scores based on data from the ERQ. The Wilcoxon *z* – *value* and *p* – *value* were calculated as 6.61 and 0.003, respectively, indicating a statistically significant improvement from *pre* to *post* measurement.

For the analysis of the SAM data, Friedman tests revealed significant differences across the three measurement phases (*pre*, *during*, *post*) in *valence* ($\chi^2(2) = 26.00, p < 0.001$), *arousal* ($\chi^2(2) = 13.17, p = 0.001$), and *dominance* ($\chi^2(2) = 12.03, p = 0.002$). Pairwise Post-hoc Wilcoxon Signed-Rank tests for *valence* showed a significant decrease from *pre* to *during* ($Z = 3.85, p < 0.001$), followed by a significant increase from *during* to *post* ($Z = 4.11, p < 0.001$), while no significant difference was observed between *pre* and *post* ($Z = 2.71, p = 0.325$). For *arousal*, a marginally significant change was observed between *pre* and *during* ($Z = 2.56, p = 0.051$), with a subsequent significant decrease from *during* to *post* ($Z = 3.81, p < 0.001$), significant difference was also observed between *pre* and *post* ($Z = 3.17, p = 0.013$). Regarding *dominance*, a significant decrease was found from *pre* to *during* ($Z = 3.21, p = 0.011$), followed by a significant increase from *during* to *post* ($Z = 2.42, p = 0.026$); however, no significant difference was noted between *pre* and *post* ($Z = 1.06, p = 0.967$).

5.2 Qualitative Findings

Three themes were constructed from our interviews, described below.

5.2.1 Emotional Transition. Participants commonly described an emotional shift, moving from feelings of stress and anxiety (especially when immersing themselves in a realistic virtual exam environment) to calmness and relaxation by the end of the session. In the *Identify your Negative Thoughts* phase, participants highlighted the realism of the virtual setting, which induced feelings of being overwhelmed and anxious (P9, P16, P21). Then, participants experienced significant emotional relief after transitioning to the *Fostering Mindful Awareness* phase. Combining soothing audio, ambient visuals, and guided meditation helped them regain control over their thoughts. As P18 described, “I guess it [*Fostering Mindful Awareness Phase*] helped me not think about the previous experience. The thoughts that came up after the exam were distracted, so that was good... The meditation kept trying to center me, making me focus on the present instead of being anxious about the future”.

5.2.2 Bubbles for Thought Management. Using bubbles as a metaphor for thoughts, particularly negative ones, was one of the central features of the MindfulReframer. Participants found physical engagement with the bubbles – grabbing, throwing, and transforming them – effective in managing their stress. For instance, one participant said, “Grabbing and throwing the bubbles felt like I could release all my stress in different ways” (P2). Similarly, P4 reported, “It felt like I could control my emotions... The bubble was soft and could float in the air, and it made me feel like I could float in the air as well. So that really calmed my stress”.

5.2.3 Thoughts Reframing. The process of CR, facilitated by the combination of mindfulness and bubble interactions, was a significant outcome of the VR experience. Participants reported increased

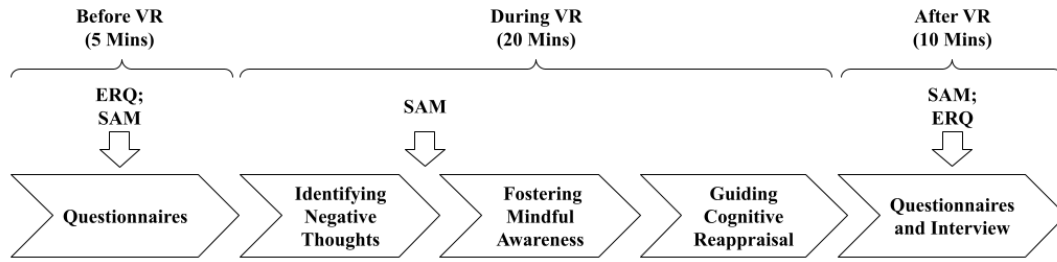


Figure 2: Experiment Procedure. The experiment consists of three main stages: before, during, and after the virtual reality experience.

awareness of their negative thought patterns, particularly those related to exam anxiety, and were able to reframe these thoughts effectively. As P10 noted, “*I could find some solutions in order to resolve my anxiety*”. The visual transformation of the bubbles was especially impactful in enhancing this process, “*When I threw the coloured bubble at the grey [bubbles], it felt like it replaced all my thoughts about the situation*” (P16). Many participants discussed how they would apply the strategies learned in VR to real life. For example, “*I found a place to sit with all my feelings and stress so I could reorganize my thoughts and get ready for the exams*” (P4). Another participant shared, “*It helped me see the exam as just one small part of life, not something that defines my future*” (P3).

6 Discussion

The present study demonstrates the potential of *MindfulReframer* to enhance CR abilities in university students for managing negative thoughts and emotions in the context of stressful situations, such as exams. In Phase I of *MindfulReframer*, our quantitative result showed that participants experienced significant decreases in valence (from positive to negative) and heightened arousal levels, indicating that the exam scenario effectively induced negative emotions. This aligns with existing research suggesting that VR can induce specific emotional responses (e.g., [20, 38]).

CR is widely recognised as a difficult skill to master, as it requires significant cognitive resources and learning ability [11, 25]. *MindfulReframer* addresses this by introducing a tangible, visual representation of negative thoughts through floating bubbles, as shown in Phases II and III. This kind of visual metaphor is also widely used in VR-based ER research [13, 30], enabling participants to manage their negative thoughts in an intuitive way. However, intense emotions can hinder the ability to develop CR strategies [1]. To overcome this, *MindfulReframer* incorporates mindfulness voice guidance, inspired by MBCT [12], to help participants distance themselves from overwhelming thoughts and emotions. This guided mindfulness process facilitates an immediate sense of calm, preparing participants to engage more effectively with reappraisal strategies, as our results have shown.

Moving to Phase III, participants actively interacted with the floating bubbles, reframing their negative thoughts in an embodied way. This process consolidated their cognitive shifts, providing a sense of control over their emotional states [25]. Furthermore, our

experience is designed to end with a visual transformation of negative thoughts into positive ones, incorporating the design concept of “tying it in” identified by Wagener et al. [39], which aims to foster acceptance and control, further enhancing the therapeutic benefits of the CR application. Our quantitative results show that after completing *MindfulReframer*, participants demonstrated significant improvement in their CR propensities, indicating that the immersive and metaphorical approach was effective in enhancing emotion regulation. This approach is grounded in MBCT’s core principles, helping individuals detach from their initial emotional reactions and enhancing cognitive flexibility, which is essential for effective CR [12]. As a novel tool for emotion regulation, *MindfulReframer* offers unique opportunities to facilitate adaptive strategies – CR strategies that have historically received less attention compared to other ER strategies, such as response-focused strategies [32].

Additionally, *MindfulReframer* provides a self-help method to develop emotion regulation strategies, addressing the limitations of traditional CR intervention that often rely on therapeutic alliances and may not transfer effectively to real-life situations [40]. By leveraging the ecological validity of VR [29], *MindfulReframer* addresses exam-related stress and has the potential to help participants apply ER strategies acquired in VR to the real world, particularly in exam-related scenarios.

7 Limitation and Future Work

Despite promising results, this study has limitations. Firstly, it relied on subjective measures, like self-reports and interviews, which may introduce bias. While important, these should be complemented by physiological measures such as heart rate variability and skin conductance to objectively assess emotional changes [18]. Secondly, the study showed short-term mood and CR improvements, but the long-term effects remain unclear. Future research should use longitudinal designs to evaluate lasting impacts. Thirdly, the ecological validity of the simulated exam scenario is limited. The environment incorporated visual cues to mimic real exam settings (e.g., a crowded classroom) and included textual stimuli about anxious thoughts, but it still relies on participants’ recollections of pre-exam anxiety to induce exam anxiety, which may not fully capture the intensity experienced in actual exam settings. Future research should evaluate *MindfulReframer* with students who are actively preparing for an exam to verify its efficacy in a naturalistic setting.

Furthermore, the current version of *MindfulReframer* focuses on mitigating negative emotions in exam scenarios. Expanding its application to other academic stressors, such as presentations and interviews, could enhance students' capacity for emotion regulation across diverse contexts. This adaptation would involve tailoring the system to context-specific scenarios. For instance, in an interview setting, the virtual environment could simulate a panel interview, curate scenario-relevant negative thoughts (e.g., every time I attend an interview, something goes wrong). Beyond academia, *Mindful-Reframer* also could be extended to workplace or social contexts, thereby broadening its utility across diverse populations.

8 Conclusion

To explore VR's potential in supporting emotion regulation, we developed *MindfulReframer*, a VR system designed to help university students manage exam-related stress through CR strategies. Quantitative and qualitative results of our study demonstrate its efficacy in enhancing CR abilities. Our work addresses a gap in VR-based emotion regulation interventions, integrating MBCT to manage exam anxiety with potential applications to broader aspects of life. We envision *MindfulReframer* as a foundation for future VR mental health interventions, encouraging further exploration of innovative psychological techniques in immersive digital environments.

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References

- [1] David H. Barlow and Todd Farchione (Eds.). 2017. *Applications of the Unified Protocol for Transdiagnostic Treatment of Emotional Disorders*. Oxford University Press, New York. <https://doi.org/10.1093/med-psych/9780190255541.001.0001>
- [2] Pedro Diniz Bernardo, Amrita Bains, Sophia Westwood, and Daniel C. Mograbi. 2020. Mood Induction Using Virtual Reality: a Systematic Review of Recent Findings. *Journal of Technology in Behavioral Science* 6, 1 (2020), 3–24. <https://doi.org/10.1007/s41347-020-00152-9>
- [3] Margaret M. Bradley and Peter J. Lang. 1994. Measuring Emotion: The Self-Assessment Manikin and the Semantic Differential. *Journal of Behavior Therapy and Experimental Psychiatry* 25, 1 (1994), 49–59.
- [4] Karen Brans, Peter Koval, Philippe Verduyn, Yan Lin Lim, and Peter Kuppens. 2013. The regulation of negative and positive affect in daily life. *Emotion* 13, 5 (2013), 926–939. <https://doi.org/10.1037/a0032400>
- [5] Alison Wood Brooks. 2014. Get excited: Reappraising pre-performance Anxiety as excitement. *Journal of Experimental Psychology: General* 143, 3 (2014), 1144–1158. <https://doi.org/10.1037/a0035325>
- [6] Laura Campbell-Sills, David H. Barlow, Timothy A. Brown, and Stefan G. Hofmann. 2006. Effects of suppression and acceptance on emotional responses of individuals with anxiety and mood disorders. *Behaviour Research and Therapy* 44, 9 (2006), 1251–1263. <https://doi.org/10.1016/j.brat.2005.10.001>
- [7] Emily Carl, Aliza T. Stein, Andrew Levihn-Coon, Jamie R. Pogue, Barbara Rothbaum, Paul Emmelkamp, Gordon J.G. Asmundson, Per Carlbring, and Mark B. Powers. 2019. Virtual reality exposure therapy for anxiety and related disorders: A meta-analysis of randomized controlled trials. *Journal of Anxiety Disorders* 61 (2019), 27–36. <https://doi.org/10.1016/j.janxdis.2018.08.003>
- [8] Desirée Colombo, Amanda Diaz-Garcia, Javier Fernandez-Álvarez, and Cristina Botella. 2021. Virtual reality for the enhancement of emotion regulation. *Clinical Psychology & Psychotherapy* 28, 3 (2021), 519–537. <https://doi.org/10.1002/cpp.2618>
- [9] Debora Cutuli. 2014. Cognitive Reappraisal and Expressive Suppression Strategies Role in the Emotion Regulation: An Overview on Their Modulatory Effects and Neural Correlates. *Frontiers in Systems Neuroscience* 8 (2014). <https://doi.org/10.3389/fnsys.2014.00175>
- [10] Thomas Ehring and Marcella L. Woud. 2023. Towards a better understanding of the role of reappraisal in psychopathology and its treatment: commentary on “Reappraising Reappraisal: An Expanded View”. *Cognition & Emotion* 37, 3 (2023), 378–383. <https://doi.org/10.1080/02699931.2023.2208341>
- [11] Brett Q. Ford, Helena R. Karnilowicz, and Iris B. Mauss. 2017. Understanding Reappraisal as a Multicomponent process: the Psychological Health Benefits of Attempting to Use Reappraisal Depend on Reappraisal success. *Emotion* 17, 6 (2017), 905–911. <https://doi.org/10.1037/emo0000310>
- [12] Eric Garland, Susan Gaylord, and Jongbae Park. 2009. The Role of Mindfulness in Positive Reappraisal. *EXPLORE* 5, 1 (2009), 37–44. <https://doi.org/10.1016/j.explore.2008.10.001>
- [13] Florian Grieger, Holger Klapperich, and Marc Hassenzahl. 2021. Trash It, Punch It, Burn It – Using Virtual Reality to Support Coping with Negative Thoughts. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 459, 6 pages. <https://doi.org/10.1145/3411763.3451738>
- [14] James J. Gross and Brett Q. Ford. 2024. *Handbook of Emotion Regulation* (3rd ed.). Guilford Press, United States.
- [15] James J. Gross and Oliver P. John. 2003. Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *Journal of Personality and Social Psychology* 85, 2 (2003), 348–362.
- [16] Melbourne Academic Mindfulness Interest Group. 2006. Mindfulness-Based Psychotherapies: A Review of Conceptual Foundations, Empirical Evidence and Practical Considerations. *Australian & New Zealand Journal of Psychiatry* 40, 4 (2006), 285–294. <https://doi.org/10.1080/j.1440-1614.2006.01794.x>
- [17] M. Guha. 2017. APA College Dictionary of Psychology (second edition). *Reference Reviews* 31, 2 (2017), 11–11. <https://doi.org/10.1108/RR-10-2016-0240>
- [18] Kunal Gupta, Sam W. T. Chan, Yun Suen Pai, Alexander Sumich, Suranga Nanayakkara, and Mark Billingham. 2021. Towards Understanding Physiological Responses to Emotional Autobiographical Memory Recall in Mobile VR Scenarios. In *Adjunct Publication of the 23rd International Conference on Mobile Human-Computer Interaction* (Toulouse & Virtual, France) (MobileHCI '21). Association for Computing Machinery, New York, NY, USA, Article 17, 5 pages. <https://doi.org/10.1145/3447527.3474864>
- [19] Jeremy P. Jamieson, Wendy Berry Mendes, Erin Blackstock, and Toni Schmader. 2010. Turning the knots in your stomach into bows: Reappraising arousal improves performance on the GRE. *Journal of Experimental Social Psychology* 46, 1 (2010), 208–212. <https://doi.org/10.1016/j.jesp.2009.08.015>
- [20] Weiwei Jiang, Maximiliane Windl, Benjamin Tag, Zhanna Sarsenbayeva, and Sven Mayer. 2024. An Immersive and Interactive VR Dataset to Elicit Emotions. *IEEE Transactions on Visualization and Computer Graphics* 30, 11 (2024), 7343–7353. <https://doi.org/10.1109/TVCG.2024.3456202>
- [21] Jon Kabat-Zinn. 2003. Mindfulness-Based Interventions in Context: Past, Present, and Future. *Clinical Psychology: Science and Practice* 10, 2 (2003), 144–156. <https://doi.org/10.1093/clipsy.bpg016>
- [22] Jon Kabat-Zinn. 2013. *Full Catastrophe living: Using the Wisdom of Your Body and Mind to Face stress, pain, and illness*. Bantam Books, New York.
- [23] Simona C. Kaplan, Amanda S. Morrison, Philippe R. Goldin, Thomas M. Olino, Richard G. Heimberg, and James J. Gross. 2017. The Cognitive Distortions Questionnaire (CD-Quest): Validation in a Sample of Adults with Social Anxiety Disorder. *Cognitive Therapy and Research* 41, 4 (2017), 576–587. <https://doi.org/10.1007/s10608-017-9838-9>
- [24] Alexandra Kitson, Alissa N Antle, and Petr Slovak. 2023. Co-designing a virtual reality intervention for supporting cognitive reappraisal skills development with youth. In *In Proceedings of the 22nd Annual ACM Interaction Design and Children Conference* (IDC '23). Association for Computing Machinery, Chicago, IL, USA, 14–26. <https://doi.org/10.1145/3585088.3589381>
- [25] Alexandra Kitson, Petr Slovak, and Alissa N. Antle. 2024. Supporting Cognitive Reappraisal With Digital Technology: A Content Analysis and Scoping Review of Challenges, Interventions, and Future Directions. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 694, 17 pages. <https://doi.org/10.1145/3613904.3642488>
- [26] Shefali Liyanage, Kiran Saqib, Amber Fozia Khan, Tijhiana Rose Thobani, Wang-Choi Tang, Cameron B. Chiarot, Bara' Abdallah AlShurman, and Zahid Ahmad Butt. 2021. Prevalence of Anxiety in University Students during the COVID-19 Pandemic: A Systematic Review. *International Journal of Environmental Research and Public Health* 19, 1 (2021), 62. <https://doi.org/10.3390/ijerph19010062>
- [27] Joana Oliveira, Susana Pedras, Richard A Inman, and Sofia Marques Ramalho. 2024. Latent profiles of emotion regulation among university students: links to repetitive negative thinking, internet addiction, and subjective wellbeing. *Frontiers in psychology* 15 (2024). <https://doi.org/10.3389/fpsyg.2024.1272643>
- [28] Catherine Nicole Marie Ortner, Mark Ste Marie, and Daniela Corno. 2016. Cognitive Costs of Reappraisal Depend on Both Emotional Stimulus Intensity and Individual Differences in Habitual Reappraisal. *PLOS ONE* 11, 12 (2016), e0167253. <https://doi.org/10.1371/journal.pone.0167253>
- [29] Thomas D. Parsons. 2015. Virtual Reality for Enhanced Ecological Validity and Experimental Control in the Clinical, Affective and Social Neurosciences. *Frontiers in Human Neuroscience* 9 (2015), 660. <https://doi.org/10.3389/fnhum.2015.00660>

- [30] Julian Rasch, Michelle Johanna Zender, Sophia Sakel, and Nadine Wagener. 2024. Mind Mansion: Exploring Metaphorical Interactions to Engage with Negative Thoughts in Virtual Reality. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 2305–2318. <https://doi.org/10.1145/3643834.3661557>
- [31] Elizabeth Seabrook, Ryan Kelly, Fiona Foley, Stephen Theiler, Neil Thomas, Greg Wadley, and Maja Nedeljkovic. 2020. Understanding How Virtual Reality Can Support Mindfulness Practice: Mixed Methods Study. *Journal of Medical Internet Research* 22, 3 (2020), e16106. <https://doi.org/10.2196/16106>
- [32] Petr Slovak, Alissa Antle, Nikki Theofanopoulou, Claudia Daudén Roquet, James Gross, and Katherine Isbister. 2023. Designing for Emotion Regulation Interventions: An Agenda for HCI Theory and Research. *ACM Trans. Comput.-Hum. Interact.* 30, 1, Article 13 (March 2023), 51 pages. <https://doi.org/10.1145/3569898>
- [33] Wally Smith, Greg Wadley, Sarah Webber, Benjamin Tag, Vassilis Kostakos, Peter Koval, and James J. Gross. 2022. Digital Emotion Regulation in Everyday Life. In *CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–15. <https://doi.org/10.1145/3491102.3517573>
- [34] Benjamin Tag, Niels van Berkel, Andrew W. Vargo, Zhanna Sarsenbayeva, Tyler Colasante, Greg Wadley, Sarah Webber, Wally Smith, Peter Koval, Tom Hollenstein, Jorge Goncalves, and Vassilis Kostakos. 2022. Impact of the global pandemic upon young people's use of technology for emotion regulation. *Computers in Human Behavior Reports* 6 (2022), 100192. <https://doi.org/10.1016/j.chbr.2022.100192>
- [35] John D. Teasdale, J. Mark G. Williams, and Zindel V. Segal. 2014. *The Mindful Way Workbook: An 8-Week Program to Free Yourself from Depression and Emotional Distress*. Guilford Press, New York.
- [36] Allison S. Troy, Anna Brunner, Amanda J. Shallcross, Rachel Friedman, and Markera C. Jones. 2018. Cognitive reappraisal and acceptance: Effects on emotion, physiology, and perceived cognitive costs. *Emotion* 18, 1 (2018), 58–74. <https://doi.org/10.1037/emo0000371>
- [37] Allison S. Troy, Amanda J. Shallcross, Tchiki S. Davis, and Iris B. Mauss. 2012. History of Mindfulness-Based Cognitive Therapy Is Associated with Increased Cognitive Reappraisal Ability. *Mindfulness* 4, 3 (2012), 213–222. <https://doi.org/10.1007/s12671-012-0114-5>
- [38] Nadine Wagener, Arne Kiesewetter, Leon Reicherts, Paweł W. Woźniak, Johannes Schöning, Yvonne Rogers, and Jasmin Niess. 2024. MoodShaper: A Virtual Reality Experience to Support Managing Negative Emotions. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 2286–2304. <https://doi.org/10.1145/3643834.3661570>
- [39] Nadine Wagener, Johannes Schöning, Yvonne Rogers, and Jasmin Niess. 2023. Letting It Go: Four Design Concepts to Support Emotion Regulation in Virtual Reality. In *Proceedings of the 2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, Shanghai, China, 763–764. <https://doi.org/10.1109/VRW58643.2023.00224>
- [40] Ya-Xin Wang and Bin Yin. 2023. A new understanding of the cognitive reappraisal technique: an extension based on the schema theory. *Frontiers in Behavioral Neuroscience* 17 (2023), 1174585. <https://doi.org/10.3389/fnbeh.2023.1174585>

A Cognitive Distortions and Corresponding Exam-related Negative Thoughts

Cognitive Distortion (defined by Kaplan et al. [23])	Exam-related Negative Thoughts
Dichotomous thinking: “I view a situation, a person or an event in ‘either-or’ terms, fitting them into only two extreme categories instead of on a continuum.”	If I don’t get a HD, I might as well have failed.
Fortune telling: “I predict the future in negative terms and believe that what will happen will be so awful that I will not be able to stand it.”	I’m going to freeze up and forget everything I’ve studied.
Discounting the positive: “I disqualify positive experiences or events insisting that they do not count.”	It doesn’t matter that I’ve been studying hard; I’ll still probably fail.
Emotional reasoning: “I believe my emotions reflect reality and let them guide my attitudes and judgments.”	I feel so anxious, this must mean I’m not prepared at all.
Labelling: “I put a fixed, global label, usually negative, on myself or others.”	I’m just a bad test-taker.
Magnification/minimisation: “I evaluate myself, others, and situations placing greater importance on the negatives and/or placing much less importance on the positives.”	That one topic I’m unsure about will probably be the main focus of the exam.
Selective abstraction: “I pay attention to one or a few details and fail to see the whole picture.”	I got one practice question wrong, so I must not understand anything.
Mind reading: “I believe that I know the thoughts or intentions of others (or that they know my thoughts or intentions) without having sufficient evidence.”	Everyone will think less of me if I fail.
Overgeneralisation: “I take isolated negative cases and generalise them, transforming them in a never-ending pattern, by repeatedly using words such as ‘always,’ ‘never,’ ‘ever,’ ‘whole,’ ‘entire,’ etc.”	Every time I take an exam, something goes wrong.
Personalisation: “I assume that others’ behaviours and external events concern (or are directed to) myself without considering other plausible explanations.”	If I fail, it will disappoint everyone who believes in me.
Should statements: “I tell myself that events, people’s behaviours, and my own attitudes ‘should’ be the way I expected them to be and not as they really are.”	I shouldn’t be feeling this anxious; I should be more confident.
Jumping to conclusions: “I draw conclusions (negative or positive) from little or no confirmatory evidence.”	I’ll likely misread a crucial question and get everything wrong.
Blaming: “I direct my attention to others as sources of my negative feelings and experiences, failing to consider my own responsibility; or, conversely, I take responsibility for others’ behaviours and attitudes.”	The professor didn’t explain things well enough, so how can I be expected to pass?
What if? “I keep asking myself questions such as “what if” something happens?”	What if I blank out and can’t remember anything I studied?
Unfair comparisons: “I compare myself with others who seem to do better than I do and place myself in a disadvantageous position.”	Everyone else seems so calm and prepared compared to me.

B Guidance in Mindfulness Awareness Phase

Becoming Aware, Gathering, Expanding Welcome to the Mindful Awareness phase. Take a moment to settle into your virtual environment. Make a deliberate and definite change in posture, so as best you can establishing now a posture that embodies a sense of dignity and wakefulness the back erect, but not stiff, dropping the shoulders, opening the chest. Let's begin our journey of mindful observation. Bringing your awareness to your inner experience, becoming aware, really aware, of your experience in this moment. so focusing your awareness inward and asking, what is my experience right now? What thoughts are here? As best you can, acknowledging thoughts as mental events, perhaps putting them into words. What feelings are here? Turning toward any sense of emotional discomfort or unpleasant feelings, acknowledging whatever feelings you find here you are. That's how it is right now. And what body sensations are you aware of right now? Perhaps quickly scanning the body to pick up any sensations of tightness or bracing you so you've got a sense of what's going on right now. Now, bring your attention to the exam-related thoughts you identified earlier. You'll see them represented as bubbles floating around you. Don't try to push them away or hold onto them. While these exam-related thoughts are present, they don't define you. They're simply mental events passing through your awareness, just like the bubbles floating in this space. As you observe these thought-bubbles, pay

attention to any emotions or physical sensations that arise. Move in close to the sense of the breath in the abdomen . . . feeling the sensations of the abdominal wall expanding as the breath comes in . . . and falling back as the breath goes out. Follow the breath all the way in and all the way out, using the breathing to anchor yourself in the present. If your mind wanders away at any time, gently guide your attention back to your breath and the floating thought-bubbles. Now expand the field of your awareness around your breathing so that it includes a sense of the body as a whole, your posture, and facial expression. If you become aware of any sensations of discomfort, tension, or resistance, take your awareness there by breathing into them on the inbreath. Then breathe out from those sensations, softening and opening with the outbreath. Remember, you're not trying to empty your mind or achieve a particular state. You're simply practising the skill of noticing your thoughts and feelings without getting caught up in them. As you continue to observe, you might notice that these thoughts and feelings change. They may intensify, fade, or transform. This is the nature of our mental experiences - they are temporary and ever-changing. Now, try to imagine that there's a part of you that's separate from these thoughts and feelings. This is your observing self - the part of you that can step back and watch your experiences without being overwhelmed by them. As best you can, take this mindful awareness with you to the next moments.